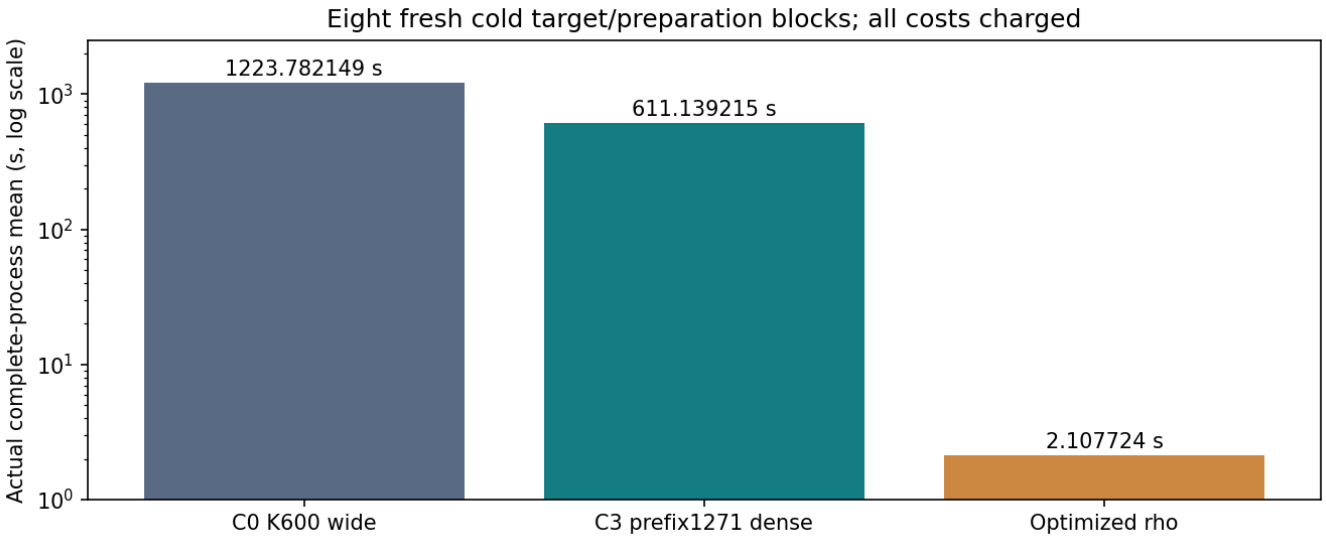
Degree-83 IC basis/setup study

Final evidence publication,16 October2026. Exact legacy a=1/tail149 degree83 curve; 53-bit prime subgroup r=8569786107849059. Question: can a better basis and relation strategy reduce preparation enough for a measured end-to-end IC win?



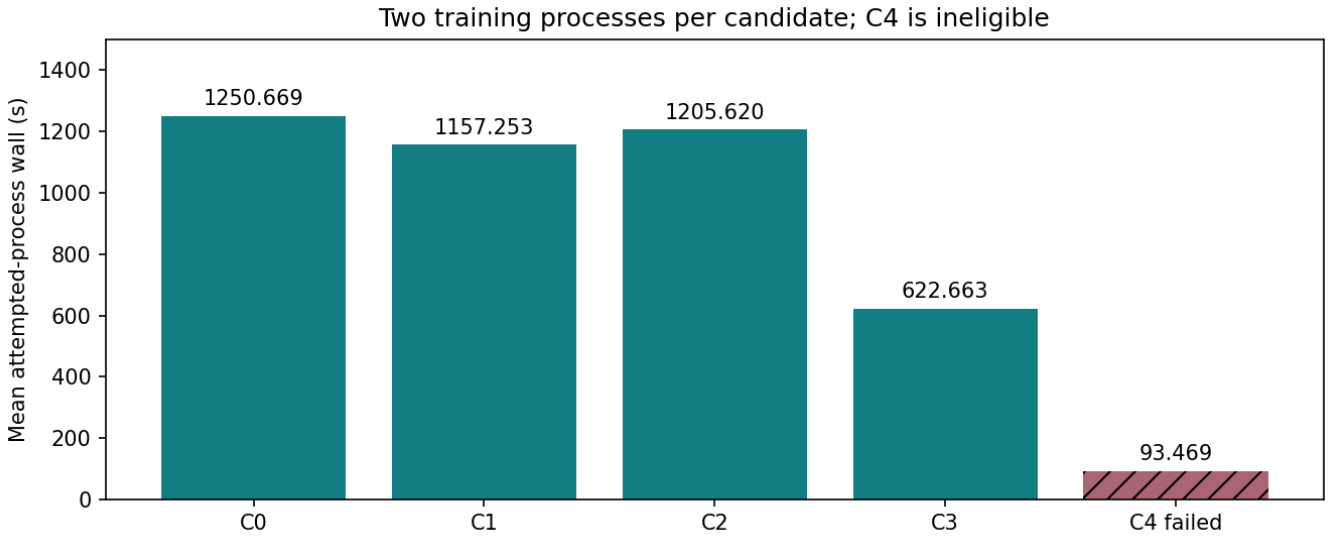
Observed cold means: C3 611.139215s; retained K600 C0 1223.782149s; optimized rho2.107724s. C3/rho289.952172, conditional95% interval[245.967290,375.396780]. C3/C0 0.499386[0.484206,0.514186]. Optimized rho has lower complete-process cost on all eight matched cold targets. All IC/rho ratios below1 favor IC.

55 attempted processes:50 completed,5 failed;92 independently verified canonical answers (72IC,20rho),30 full rank/log transcripts,5 identical rho A/A work pairs. Planned118 answers were not achieved. Three8-target rho processes rejected the input contract; no cohort IC/rho ratio or winner is available. The two C4 training allocation failures also remain in the evidence.

This addendum closes publication of the attempted bounded campaign and explicitly retains the incomplete cohort comparison. Historical87 pages are preserved in the appended comprehensive report. PR1899; required CI and human review remain merge gates.

Prospective design and basis selection

Candidate	Mathematical basis	Representation
C0	Retained K600	Wide control
C1	Identical retained K600	Dense table
C2	Next600 orbits after first600	Dense table
C3	Public-scan prefix1271	Dense table
C4	Public-scan prefix1798	Dense table



C0/C1 isolates representation; C1/C2 isolates point selection. C3/C4 enlarge the prefix. Dense packing is engineering, not a new mathematical basis. The first600 draft was exactly K600 and was corrected to a disjoint window before any timing. C2 charges construction of1200 then removal/reindexing of600. No pre-solved logs are supplied.

Training uses two public targets outside1911 excluded signed orbits, rank seeds29/43 matched across bases. Selection is arithmetic mean whole-process wall over both full-rank-verified runs; failed candidates are ineligible and ties lexical. C3 selected at622.6626231075s versus C0 1250.669494426s. Selection committed19664f02f before generating eight holdouts outside1913 excluded signed orbits. Worker/protocol freeze e07da4efd; holdout matrix/controller freeze1e93e4ca1.

Matched cold results and uncertainty

Exact polynomial: z^83+z^7+z^4+z^2+1, a=b=1; G=(661424465139818486029201 5,886826041002924939153952 9), lambda=2998004823950457, signed Frobenius orbit166. Slug icv1-f2m83-t6151469093347-9eba6cd5. No tail45 or a=0 substitution.

Ratio of native mean costs	Estimate	Conditional95% interval
C3/rho	289.952172	[245.967290, 375.396780]
C0/rho	580.617776	[489.180227, 758.610292]
C3/C0	0.499386	[0.484206, 0.514186]

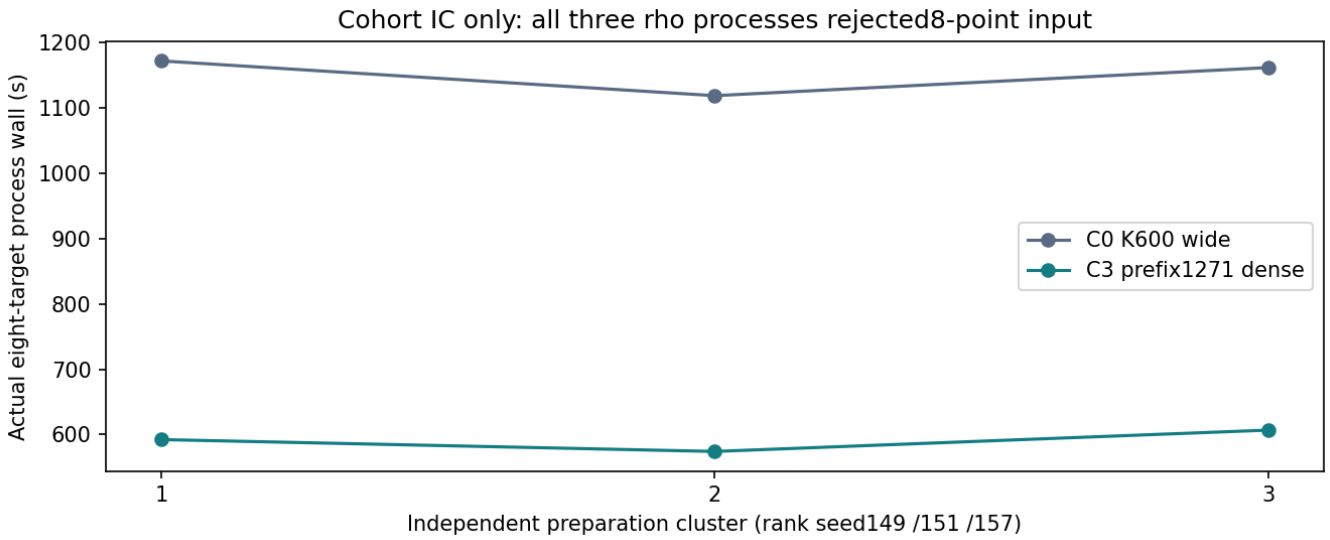
All26 cold processes/answers passed;16 full IC rank/log audits and one A/A pair passed with zero invalid/missing. Primary paired bootstrap resamples eight matched target/preparation blocks,20,000 draws, seed202610157801. Quantiles use conventional midpoint median and type7 linear interpolation. These are conditional intervals for this panel on one guest, not independent-host guarantees.

Both methods charge actual process launch, loading, preparation, solve, generic scalar verification and output. IC original materialized-K600 selection/construction and OS-cache flushing remain unpriced. C3 constructs its public-scan base in every process. The native CPU3 reservation does not make the physical host exclusive; PSI unavailable and unmovable daemons are logged. No calibrated S or L2 admission is claimed.

Aligned query clocks match: before generic public-Q curve/subgroup validation through solve/recovery, generic scalar replay and answer serialization/write/flush. Both exclude input-file parsing and their own receipt emission. The cold aligned-query mean C3/rho ratio is 0.115579; this is secondary and does not remove preparation from the end-to-end finding.

No old IC times are paired with new rho times, no gains are multiplied across studies, and no projected1685-target experiment is performed. The old32-target matched cohort and this failed8-target cohort remain separate historical regimes.

Final cohorts: results and failed reference



Three genuinely reused8-target IC preparation clusters completed. C0 mean1150.628970s; C3 mean591.192669s. Native C3/C0 0.513800 [0.505576, 0.522482],20,000 paired cluster draws, seed202610157802. Only three preparation clusters limit uncertainty. This IC-basis comparison is not an IC/rho comparison.

All three rho processes returned2 in about0.006s before any answer: inherited worker lines107-108 accept only a singleton or exactly32 points. The predeclared8-point matrix and frozen worker were incompatible. This implementation mismatch should have been caught before timing. No solves are retried; raw records, failed audit and analyzer refusal remain unchanged. Failed-launch wall time is never treated as rho solve time.

Six cohort IC rank/log audits and the final rho A/A pair pass:8/11 processes and50/74 canonical answers, including48IC and2rho. The full-stage analyzer correctly refuses to generate complete cohort statistics. Supplemental native analysis reports the six verified IC processes only and leaves every rho field null. See problems/n83-eight-target-rho-contract.md for the unresolved acceptance conditions.

All cold target aligned-clock rows

Target	C0 query(s)	C3 query(s)	Rho query(s)	C3/rho
0	0.932516	0.219019	2.156455	0.101564
1	0.449062	0.142074	2.270929	0.062562
2	5.546242	0.082767	0.489860	0.168961
3	0.461892	0.435195	2.071393	0.210097
4	0.750041	0.103484	1.969131	0.052553
5	1.268562	0.216246	2.545347	0.084957
6	0.467315	0.268216	2.177855	0.123156
7	0.971125	0.269555	1.343932	0.200572

Each row is one fresh public target, one complete preparation per IC method, and one independent rho solve. Clock boundaries match in both methods. These are native recorded intervals; native ratios of mean query costs are secondary. Full-process target rows and exact public-Q identities are in the matrix, sidecars and raw answers. Index0..7 follows the frozen holdout target order.

Main cold rho retains253 internal capped walks, all charged. Capped walks that later recover are distinct from failed processes; no slow cases or internal restarts are dropped. All scalar answers are canonical and independently replayed against the exact public Q.

All cohort targets: three recorded repeats

Target / role	Repeat1(s)	Repeat2(s)	Repeat3(s)	Native mean(s)
0 / control	0.944780	0.893198	1.048826	0.962268
0 / selected	0.212327	0.231449	0.218369	0.220715
1 / control	0.423121	0.416658	0.452766	0.430849
1 / selected	0.134075	0.146941	0.139594	0.140203
2 / control	5.111935	5.334300	5.495097	5.313777
2 / selected	0.082102	0.104607	0.085545	0.090751
3 / control	0.486521	0.420921	0.528880	0.478774
3 / selected	0.382984	0.430788	0.397981	0.403918
4 / control	0.734322	0.816188	0.833977	0.794829
4 / selected	0.093264	0.099044	0.097516	0.096608
5 / control	1.249948	1.363634	1.210447	1.274676
5 / selected	0.198137	0.205818	0.204467	0.202807
6 / control	0.415989	0.437810	0.408795	0.420865
6 / selected	0.232286	0.241682	0.235639	0.236535
7 / control	0.860173	0.932045	0.868460	0.886893
7 / selected	0.255603	0.270130	0.263494	0.263076

Every target has three verified IC query intervals under independent preparations. Rho repeats, online ratios and intervals are unavailable: all three cohort rho workers rejected the8-point input before answering. There is no substitution from cold or historical32-target rho measurements.

Rank seeds149/151/157; target rotations0/3/5, reversal in cohort2 as numbered by protocol. Exact order and per-Q rho seeds are in the frozen matrix. No target-dependent rho DP/walk state sharing was introduced. Repeated-target rows remain secondary to actual complete-process costs.

Complete-process wall distributions

Stage / role	n	Mean	Median	p90	p95	Max
cohort/control	3	1150.62897	1161.55340	1169.71750	1170.73801	1171.75852
cohort/selected	3	591.19267	592.41343	603.99544	605.44319	606.89095
cold/control	8	1223.78215	1235.83771	1287.73823	1289.73791	1291.73758
cold/rho	8	2.10772	2.35845	2.57344	2.66619	2.75894
cold/selected	8	611.13922	614.17588	629.22241	632.25111	635.27981
training/C0	2	1250.66949	1250.66949	1295.31947	1300.90072	1306.48196
training/C1	2	1157.25336	1157.25336	1160.78659	1161.22824	1161.66989
training/C2	2	1205.62014	1205.62014	1208.00149	1208.29916	1208.59682
training/C3	2	622.66262	622.66262	625.45860	625.80810	626.15760
training/rho	2	2.60835	2.60835	3.64864	3.77868	3.90871

Units: seconds. Native arithmetic means, midpoint medians and type7 linear-interpolated p90/p95. Failed C4 and cohort-rho attempts are excluded from successful-cost distributions and retained individually in the attempt tables. A/A controls have their own spread table. No complete cohort-rho distribution exists.

Whole-process CPU is user+system from wait4; peak RSS is whole-process maximum. Endpoint memory samples below are diagnostic and cannot establish exclusive phase peaks. Memory-limit-hit flag is null. Online distributions pool recorded query intervals as descriptive values; they do not create additional independent preparation clusters.

Complete-process CPU distributions

Stage / role	n	Mean	Median	p90	p95	Max
cohort/control	3	1150.53498	1161.46473	1169.61113	1170.62943	1171.64773
cohort/selected	3	591.13565	592.35520	603.93636	605.38401	606.83165
cold/control	8	1223.69038	1235.74398	1287.64698	1289.65669	1291.66639
cold/rho	8	2.10268	2.35091	2.56957	2.66191	2.75425
cold/selected	8	611.07954	614.11118	629.16131	632.19175	635.22219
training/C0	2	1250.57687	1250.57687	1295.23139	1300.81321	1306.39502
training/C1	2	1157.12244	1157.12244	1160.62669	1161.06472	1161.50275
training/C2	2	1205.51856	1205.51856	1207.90047	1208.19821	1208.49595
training/C3	2	622.60901	622.60901	625.40426	625.75367	626.10307
training/rho	2	2.60311	2.60311	3.64153	3.77134	3.90114

Units: seconds. Native arithmetic means, midpoint medians and type7 linear-interpolated p90/p95. Failed C4 and cohort-rho attempts are excluded from successful-cost distributions and retained individually in the attempt tables. A/A controls have their own spread table. No complete cohort-rho distribution exists.

Whole-process CPU is user+system from wait4; peak RSS is whole-process maximum. Endpoint memory samples below are diagnostic and cannot establish exclusive phase peaks. Memory-limit-hit flag is null. Online distributions pool recorded query intervals as descriptive values; they do not create additional independent preparation clusters.

Whole-process peak RSS distributions

Stage / role	n	Mean	Median	p90	p95	Max
cohort/control	3	5834199040	5834178560	5834227712	5834233856	5834240000
cohort/selected	3	10832474112	10832515072	10832521625	10832522444	10832523264
cold/control	8	5834199040	5834207232	5834242457	5834245324	5834248192
cold/rho	8	31169536	36505600	36527308	36550246	36573184
cold/selected	8	10832451072	10832455680	10832520396	10832521830	10832523264
training/C0	2	5834151936	5834151936	5834225664	5834234880	5834244096
training/C1	2	2613014528	2613014528	2613014528	2613014528	2613014528
training/C2	2	2563819520	2563819520	2563844096	2563847168	2563850240
training/C3	2	10832486400	10832486400	10832519168	10832523264	10832527360
training/rho	2	27981824	27981824	34802483	35655065	36507648

Units: bytes. Native arithmetic means, midpoint medians and type7 linear-interpolated p90/p95. Failed C4 and cohort-rho attempts are excluded from successful-cost distributions and retained individually in the attempt tables. A/A controls have their own spread table. No complete cohort-rho distribution exists.

Whole-process CPU is user+system from wait4; peak RSS is whole-process maximum. Endpoint memory samples below are diagnostic and cannot establish exclusive phase peaks. Memory-limit-hit flag is null. Online distributions pool recorded query intervals as descriptive values; they do not create additional independent preparation clusters.

Aligned-query distributions (secondary)

Stage / role	n	Mean	Median	p90	p95	Max
cohort/control	24	1.32037	0.84707	3.98744	5.30095	5.49510
cohort/selected	24	0.20683	0.20907	0.34913	0.39573	0.43079
cold/control	8	1.35584	0.84128	2.55187	4.04905	5.54624
cold/rho	8	1.87811	2.11392	2.35325	2.44930	2.54535
cold/selected	8	0.21707	0.21763	0.31925	0.37722	0.43519
training/C0	2	0.46827	0.46827	0.78973	0.82992	0.87010
training/C1	2	0.50794	0.50794	0.86298	0.90736	0.95174
training/C2	2	0.70781	0.70781	1.00046	1.03704	1.07362
training/C3	2	0.31775	0.31775	0.34619	0.34975	0.35330
training/rho	2	2.39750	2.39750	3.43700	3.56693	3.69687

Units: seconds. Native arithmetic means, midpoint medians and type7 linear-interpolated p90/p95. Failed C4 and cohort-rho attempts are excluded from successful-cost distributions and retained individually in the attempt tables. A/A controls have their own spread table. No complete cohort-rho distribution exists.

Whole-process CPU is user+system from wait4; peak RSS is whole-process maximum. Endpoint memory samples below are diagnostic and cannot establish exclusive phase peaks. Memory-limit-hit flag is null. Online distributions pool recorded query intervals as descriptive values; they do not create additional independent preparation clusters.

Training attempted processes 1-15

Process	Status	Wall(s)	CPU(s)	PeakRSS(bytes)
aa_0_0	complete	4.630801	4.625734	62119936
aa_0_1	complete	4.764793	4.686541	62115840
aa_1_0	complete	2.018198	2.011767	19398656
aa_1_1	complete	2.085859	2.081003	19464192
train_0_C0	complete	1306.481963	1306.395022	5834244096
train_0_C1	complete	1161.669895	1161.502747	2613014528
train_0_C2	complete	1208.596824	1208.495951	2563850240
train_0_C3	complete	626.157598	626.103073	10832527360
train_0_C4	FAILED	93.587788	93.559610	6588936192
train_0_rho	complete	1.307987	1.305073	19456000
train_1_C4	FAILED	93.349774	93.330215	6588801024
train_1_C3	complete	619.167648	619.114956	10832445440
train_1_C2	complete	1202.643457	1202.541160	2563788800
train_1_C1	complete	1152.836831	1152.742143	2613014528
train_1_C0	complete	1194.857025	1194.758710	5834059776

Every declared process appears, including failures, internal-cap costs and controls. Values copied from native final aggregation; execution and raw hashes bind the records. Failed process times are attempted costs, never successful solve observations.

Both C4 processes failed allocation of12884901888 bytes under the common12GiB address-space envelope; partial traces and outputs are preserved. C4 is ineligible. This wide-state allocation policy failure does not prove larger bases or alternative packed-state layouts impossible.

Training attempted processes 16-18

Process	Status	Wall(s)	CPU(s)	PeakRSS(bytes)
train_1_rho	complete	3.908712	3.901139	36507648
aa_2_0	complete	4.435090	4.429189	62103552
aa_2_1	complete	4.296349	4.290989	62119936

Every declared process appears, including failures, internal-cap costs and controls. Values copied from native final aggregation; execution and raw hashes bind the records. Failed process times are attempted costs, never successful solve observations.

Both C4 processes failed allocation of12884901888 bytes under the common12GiB address-space envelope; partial traces and outputs are preserved. C4 is ineligible. This wide-state allocation policy failure does not prove larger bases or alternative packed-state layouts impossible.

Cold attempted processes 1-15

Process	Status	Wall(s)	CPU(s)	PeakRSS(bytes)
cold_0_control	complete	1291.737583	1291.666394	5834174464
cold_0_selected	complete	591.732233	591.685156	10832461824
cold_0_rho	complete	2.398398	2.391504	36507648
cold_1_selected	complete	618.685768	618.631645	10832388096
cold_1_rho	complete	2.493933	2.490421	36507648
cold_1_control	complete	1172.716256	1172.629776	5834248192
cold_2_rho	complete	0.714578	0.713094	10948608
cold_2_control	complete	1268.694905	1268.599469	5834166272
cold_2_selected	complete	617.671766	617.612829	10832490496
cold_3_control	complete	1286.024217	1285.924372	5834240000
cold_3_rho	complete	2.318505	2.310324	36573184
cold_3_selected	complete	635.279809	635.222192	10832519168
cold_4_rho	complete	2.194436	2.190663	36368384
cold_4_selected	complete	584.751779	584.683521	10832523264
cold_4_control	complete	1164.867769	1164.756421	5834240000

Every declared process appears, including failures, internal-cap costs and controls. Values copied from native final aggregation; execution and raw hashes bind the records. Failed process times are attempted costs, never successful solve observations.

Cold attempted processes 16-26

Process	Status	Wall(s)	CPU(s)	PeakRSS(bytes)
cold_5_selected	complete	610.679993	610.609521	10832388096
cold_5_control	complete	1230.493906	1230.398680	5834240000
cold_5_rho	complete	2.758940	2.754250	36503552
cold_6_control	complete	1134.541037	1134.458637	5834108928
cold_6_selected	complete	603.685988	603.627646	10832388096
cold_6_rho	complete	2.399296	2.392685	36507648
cold_7_selected	complete	626.626385	626.563787	10832449536
cold_7_rho	complete	1.583708	1.578464	19439616
cold_7_control	complete	1241.181517	1241.089280	5834174464
aa_3_0	complete	1.842876	1.840532	19468288
aa_3_1	complete	1.862018	1.856345	19464192

Every declared process appears, including failures, internal-cap costs and controls. Values copied from native final aggregation; execution and raw hashes bind the records. Failed process times are attempted costs, never successful solve observations.

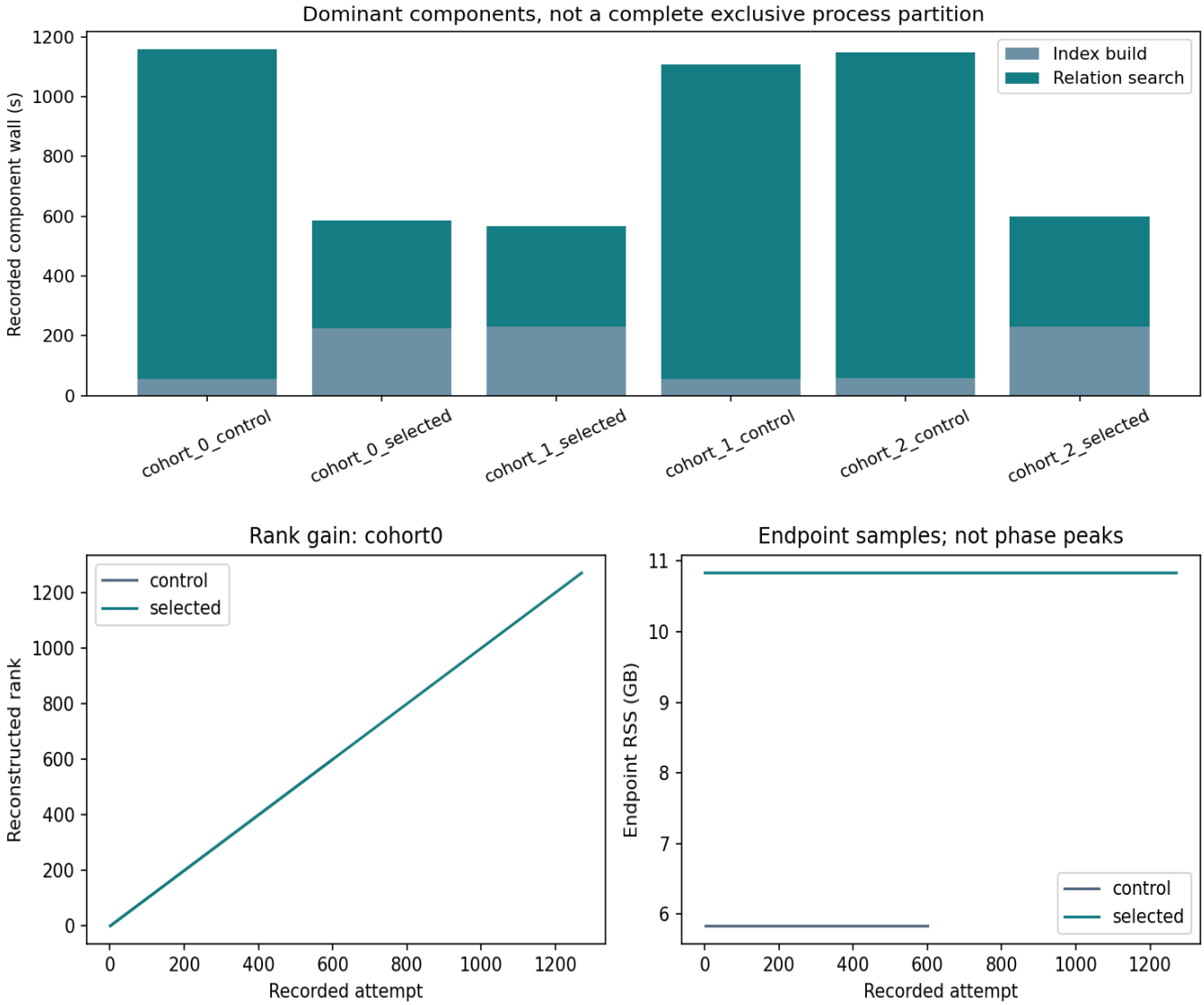
Cohort attempted processes 1-11

Process	Status	Wall(s)	CPU(s)	PeakRSS(bytes)
cohort_0_control	complete	1171.758520	1171.647730	5834178560
cohort_0_selected	complete	592.413430	592.355203	10832384000
cohort_0_rho	FAILED	0.005649	0.001023	1900544
cohort_1_selected	complete	574.273632	574.220087	10832523264
cohort_1_rho	FAILED	0.005605	0.000923	1900544
cohort_1_control	complete	1118.574987	1118.492481	5834240000
cohort_2_rho	FAILED	0.005551	0.001011	1892352
cohort_2_control	complete	1161.553402	1161.464725	5834178560
cohort_2_selected	complete	606.890945	606.831654	10832515072
aa_4_0	complete	1.582627	1.574619	19464192
aa_4_1	complete	1.541704	1.538604	19378176

Every declared process appears, including failures, internal-cap costs and controls. Values copied from native final aggregation; execution and raw hashes bind the records. Failed process times are attempted costs, never successful solve observations.

All three rho entries returned2 without answers. Their tiny launch costs cannot be compared with IC solve costs. This stage has no complete matched IC/rho result. The final two singleton rho A/A controls did complete and have identical work.

Where preparation time and memory go



For all30 completed IC processes, native profiles export every attempt: probes, states, found/useful relation, rank before/after and gain, exclusive anchor/search/check/row/elimination times and endpoint RSS/HWM. Backsolve and prior trace formatting/write/flush overhead are recorded separately. Producer rank/process residuals include uncalibrated instrumentation; these panels are not a complete exclusive cost partition.

Native rank/setup profiles 1-10

Process	Attempts / gain	Probes	Index(s)	Search(s)	Elim(s)	Backsolve(s)
train_0_C0	600 / 600	1437855278	69.991	1234.457	0.1355	0.001014
train_0_C1	600 / 600	1437855278	53.948	1105.872	0.1257	0.000974
train_0_C2	600 / 600	1472146321	55.042	1150.648	0.1318	0.000917
train_0_C3	1271 / 1271	507472060	249.491	371.263	1.0671	0.004008
train_1_C3	1271 / 1271	507465286	239.943	373.928	1.0524	0.003947
train_1_C2	600 / 600	1477593822	55.243	1145.211	0.1250	0.001061
train_1_C1	600 / 600	1340076576	59.172	1090.793	0.1288	0.001075
train_1_C0	600 / 600	1340076576	61.566	1130.508	0.1276	0.001222
cold_0_control	600 / 600	1508460462	59.910	1228.993	0.1398	0.000881
cold_0_selected	1271 / 1271	486341850	233.538	352.898	1.0579	0.003945

Index and relation search dominate these recorded preparation profiles; elimination and backsolve are much smaller. Every useful relation/rank gain is independently reconstructed. Probe/state/time counters are producer observations, not independently repeated search timings. Per-attempt details, all other exclusive phases and memory curves are in native_final.json/profiles and immutable rank.jsonl.

Process	State layout / slot	State bytes	Root-table bytes
train_0_C0	wide_u128_roots / 48	1610612736	4294967296
train_0_C1	wide_u128_roots / 48	1610612736	1073741824
train_0_C2	wide_u128_roots / 48	1610612736	1073741824
train_0_C3	wide_u128_roots / 48	6442450944	4294967296
train_1_C3	wide_u128_roots / 48	6442450944	4294967296
train_1_C2	wide_u128_roots / 48	1610612736	1073741824
train_1_C1	wide_u128_roots / 48	1610612736	1073741824
train_1_C0	wide_u128_roots / 48	1610612736	4294967296
cold_0_control	wide_u128_roots / 48	1610612736	4294967296
cold_0_selected	wide_u128_roots / 48	6442450944	4294967296

C3 uses packed_u128 root table slots16 bytes with4GiB table allocation and wide_u128_roots states with6GiB allocation. Retained memory policy is recorded, not a claim of optimal packing. Original K600 construction, OS-cache flushing and instrumentation calibration remain outside the priced boundary.

Native rank/setup profiles 11-20

Process	Attempts / gain	Probes	Index(s)	Search(s)	Elim(s)	Backsolve(s)
cold_1_selected	1271 / 1271	503328548	239.091	374.424	1.0597	0.004673
cold_1_control	600 / 600	1348839196	60.659	1109.787	0.1088	0.000927
cold_2_control	600 / 600	1479469665	62.286	1198.939	0.1375	0.000922
cold_2_selected	1271 / 1271	517897387	239.143	373.453	1.0649	0.004440
cold_3_control	600 / 600	1516538656	60.651	1223.075	0.1364	0.000852
cold_3_selected	1271 / 1271	539276639	236.923	392.661	1.1267	0.005878
cold_4_selected	1271 / 1271	471262617	238.170	341.540	1.0264	0.004259
cold_4_control	600 / 600	1402530634	59.283	1103.013	0.1231	0.000871
cold_5_selected	1271 / 1271	531392681	231.933	373.645	1.0881	0.004209
cold_5_control	600 / 600	1499148573	58.336	1169.024	0.1412	0.000902

Index and relation search dominate these recorded preparation profiles; elimination and backsolve are much smaller. Every useful relation/rank gain is independently reconstructed. Probe/state/time counters are producer observations, not independently repeated search timings. Per-attempt details, all other exclusive phases and memory curves are in native_final.json/profiles and immutable rank.jsonl.

Process	State layout / slot	State bytes	Root-table bytes
cold_1_selected	wide_u128_roots / 48	6442450944	4294967296
cold_1_control	wide_u128_roots / 48	1610612736	4294967296
cold_2_control	wide_u128_roots / 48	1610612736	4294967296
cold_2_selected	wide_u128_roots / 48	6442450944	4294967296
cold_3_control	wide_u128_roots / 48	1610612736	4294967296
cold_3_selected	wide_u128_roots / 48	6442450944	4294967296
cold_4_selected	wide_u128_roots / 48	6442450944	4294967296
cold_4_control	wide_u128_roots / 48	1610612736	4294967296
cold_5_selected	wide_u128_roots / 48	6442450944	4294967296
cold_5_control	wide_u128_roots / 48	1610612736	4294967296

C3 uses packed_u128 root table slots16 bytes with4GiB table allocation and wide_u128_roots states with6GiB allocation. Retained memory policy is recorded, not a claim of optimal packing. Original K600 construction, OS-cache flushing and instrumentation calibration remain outside the priced boundary.

Native rank/setup profiles 21-30

Process	Attempts / gain	Probes	Index(s)	Search(s)	Elim(s)	Backsolve(s)
cold_6_control	600 / 600	1380993645	55.943	1076.306	0.1145	0.000973
cold_6_selected	1271 / 1271	526140768	228.645	369.758	1.1198	0.004171
cold_7_selected	1271 / 1271	538488695	235.602	385.675	1.1134	0.004457
cold_7_control	600 / 600	1512720710	57.520	1180.977	0.1121	0.000848
cohort_0_control	600 / 600	1440607786	55.915	1103.708	0.1316	0.001024
cohort_0_selected	1271 / 1271	525159939	224.590	361.423	1.0333	0.004287
cohort_1_selected	1271 / 1271	491280327	229.726	337.722	1.0071	0.004499
cohort_1_control	600 / 600	1373542914	56.864	1049.319	0.1310	0.000919
cohort_2_control	600 / 600	1417201965	57.702	1091.149	0.1279	0.000865
cohort_2_selected	1271 / 1271	525711233	231.206	368.996	1.0740	0.004721

Index and relation search dominate these recorded preparation profiles; elimination and backsolve are much smaller. Every useful relation/rank gain is independently reconstructed. Probe/state/time counters are producer observations, not independently repeated search timings. Per-attempt details, all other exclusive phases and memory curves are in native_final.json/profiles and immutable rank.jsonl.

Process	State layout / slot	State bytes	Root-table bytes
cold_6_control	wide_u128_roots / 48	1610612736	4294967296
cold_6_selected	wide_u128_roots / 48	6442450944	4294967296
cold_7_selected	wide_u128_roots / 48	6442450944	4294967296
cold_7_control	wide_u128_roots / 48	1610612736	4294967296
cohort_0_control	wide_u128_roots / 48	1610612736	4294967296
cohort_0_selected	wide_u128_roots / 48	6442450944	4294967296
cohort_1_selected	wide_u128_roots / 48	6442450944	4294967296
cohort_1_control	wide_u128_roots / 48	1610612736	4294967296
cohort_2_control	wide_u128_roots / 48	1610612736	4294967296
cohort_2_selected	wide_u128_roots / 48	6442450944	4294967296

C3 uses packed_u128 root table slots16 bytes with4GiB table allocation and wide_u128_roots states with6GiB allocation. Retained memory policy is recorded, not a claim of optimal packing. Original K600 construction, OS-cache flushing and instrumentation calibration remain outside the priced boundary.

Independent audits and rho A/A controls

Pair / stage	First wall(s)	Second wall(s)	Larger / smaller	Work
aa_0 / training	4.630801	4.764793	1.028935	identical
aa_1 / training	2.018198	2.085859	1.033526	identical
aa_2 / training	4.435090	4.296349	1.032293	identical
aa_3 / cold	1.842876	1.862018	1.010387	identical
aa_4 / cohort	1.582627	1.541704	1.026543	identical

Native recursion compares every deterministic A/A outcome field; clocks need not match. Five pairs span training, cold and cohort phases. Independent rank checker costs are excluded from solver process costs and separately wait4-metered; per-check wall/user/system CPU/peakRSS and logs are in training-audit, cold-audit and cohort-audit. No audit was repeated to replace a failed result.

Full audits independently reconstruct every modular relation row and rank gain, backsolve all base logs, check every representative[log]G, anchored target/four-point witness/intermediate and final answer/log reconstruction. Training selection independently checked all16 canonical answers; cold26 and cohort50 were independently checked. Thirty transcripts cover72 IC answers, and all20 available rho answers pass canonical scalar replay.

The cohort checker initially reports “Cannot read input file” for missing answer files. Raw process receipts and stderr establish why those files are absent. That diagnostic is retained unchanged; no checker correction is needed. The full-stage statistical gate refuses incomplete evidence as intended. The new final aggregator binds all attempts, raw hashes and previous audit/selection seals, recursively checks all five A/A pairs, and emits null for the missing batch comparison.

Scope, acceptance and unresolved obligations

Requirement	Actual evidence / status
Basis diversity and training-only selection	Verified C0..C3; C4 failed allocations retained
Fresh matched cold complete costs	Verified8 triplets; rho lower cost
Rank/log export and independent audit	Verified30 full transcripts /72IC answers
Three8-target matched cohorts	PARTIAL: sixIC and AA passed; three rho failed
Full actual accounting	55 attempts /50complete /5failed /92 answers
Unqualified IC-over-rho winner	Not established; cold ratio is above1
New solve / projected1685-target run	Not performed
Independent host, calibrated S, L2	Unset
Current CI and required human review	Merge gates, not scientific admission

One guest, three shared preparation clusters, physical host not exclusive. Isolation records and unavailable PSI/unmovable daemons are preserved. No population-wide, global-optimum, exponent, asymptotic or deployed-security conclusion follows from this one field and subgroup. Larger-base feasibility under other layouts remains open.

The missing batch reference is an implementation/workload-contract gap. A separately versioned prospective correction would need positive8-target native acceptance and duplicate/malformed tests before freezing and timing. No replacement campaign is run in this task. This publication preserves the attempted study; it does not mark the failed matched-cohort acceptance requirement complete.

Source and evidence map

PROTOCOL.md — immutable prospective parameters and acceptance

provenance/SOURCE_SHA256SUMS and BINARY_SHA256SUMS — training identities

provenance/holdout/matrix.json and FREEZE_SHA256SUMS — remaining37 inputs/controller

selection/selection_freeze.json — pre-holdout C3 selection

records/training-campaign, cold-campaign, cohort-campaign — every raw process

records/training-audit, cold-audit, cohort-audit — independent receipts/meters

reports/run-20261015T1516Z — inspected4-page training stage PDF

reports/run-20261015T1842Z — inspected4-page cold stage PDF/native statistics

reports/run-20261015T2311Z — cohort stage report and failed receipt

postprocessing/final_aggregate.cpp — new native binding/profile/statistics export

postprocessing/test_final_aggregate.cpp — coverage/A-A/SHA tamper guards

checks/final — frozen-source seals, native checks, final PDF QA/preservation

reports/final-20261016/native_final.json — source-linked values shown here

delivery — archive manifest, reconstruction paths and CRC/full-hash receipt

problems/n83-eight-target-rho-contract.md — unresolved reference obligation

Study paths relative to research/n83_basis_setup_20261015 except problems/. Reports, editable figures, SVG and150dpiPNG copies accompany benchmark index20261015-n83-basis-setup. Canonical diagnostic changes preserve historical entries and do not promote this result into admitted S/floor/ledger rankings. PR https://github.com/aburan28/crypto/pull/1899 remains dependent on parent1828 until integration. No reviewer is requested without the owner naming one; required human review including Cursor remains a gate.

Python only formats recorded native values, draws figures, renders/inspects PDF pages and packages existing files. No scientific result or experimental logic is recomputed in this presentation. The comprehensive file appends this standalone addendum to the unchanged87-page report (SHAaf881641988a98bec3177e286694233f7b03731e82b6517a51ea78fc5e19793a).